

Notebook Nomenclature & Data Consistency Audit Report

Date: 2026-03-20
Reference: DEC-012 (P0 Nomenclature Standardization)

Correct Nomenclature

Code	Name	Description
P0	Spatially-stratified uniform	Latitude-based spatial selection, 1 unit each (canonical baseline)
P1	Demand-proportional	Units allocated proportional to precinct demand rates
P2	Demand-weighted optimized	MIP minimizing demand-weighted expected response time

Data source: `results/simulation/production/` (contains spatial P0 results, ~3.17 min at K=20)

Notebooks Audited (18 total)

Core Notebooks (9 files)

Notebook	Changes	Details
01_end_to_end_workflow.ipynb	Cleared outputs	33 stale output cells cleared (already used correct code with <code>spatially_stratified_allocation</code> and <code>production_v2</code>)
02_eda_spatiotemporal.ipynb	Cleared outputs	20 output cells cleared
03_input_modeling.ipynb	No changes	Already consistent
04_service_travel_proxy.ipynb	Cleared outputs	11 output cells cleared
05_optimization.ipynb	Fixed nomenclature	P0 descriptions changed from “Uniform” to “Spatial Baseline”; 5 markdown cells updated
06_simulation_debug.ipynb	Fixed code + nomenclature	<code>P0_uniform</code> → <code>P0</code> in JSON key references; section titles and labels updated; 6 output cells cleared
07_production_results.ipynb	Cleared stale outputs	16 output cells cleared (were showing old P0 metrics: 8.08 min, 64.4% coverage)
08_statistical_analysis.ipynb	Fixed metrics + cleared outputs	Updated stale metric references (64.2% → ~99.7%, 68% → ~19% reduction); 14 output cells cleared
09_cbd_analysis.ipynb	No changes	Already consistent

Colab Notebooks (9 files)

Notebook	Changes	Details
00_colab_setup_and_data.ipynb	Fixed imports	<code>uniform_allocation</code> → <code>spatially_stratified_allocation</code> in import verification
01_colab_eda_spatiotemporal.ipynb	No changes	Already consistent
02_colab_demand_modeling.ipynb	No changes	Already consistent
03_colab_service_modeling.ipynb	No changes	Already consistent
04_colab_optimization.ipynb	Fixed code + nomenclature	<code>uniform_allocation</code> → <code>spatially_stratified_allocation</code> ; P0 section header and summary updated
05_colab_simulation.ipynb	Fixed code	Fallback allocation generation uses <code>spatially_stratified_allocation</code>
06_colab_statistical_analysis.ipynb	No changes	Already consistent
07_colab_visualization_reporting.ipynb	No changes	Already had correct labels (P0 (Spatial Baseline))
EMS_Optimization_Complete_Pipeline.ipynb	Fixed code + labels	<code>uniform_allocation</code> → <code>spatially_stratified_allocation</code> ; P0 label updated to “Spatial Baseline”

Summary of Changes

Category	Count
Code fixes (<code>uniform_allocation</code> → <code>spatially_stratified_allocation</code>)	5 notebooks
JSON key fixes (<code>P0_uniform</code> → <code>P0</code>)	1 notebook
Label/description fixes (Uniform → Spatial Baseline)	3 notebooks
Stale metric corrections (64.2% → ~99.7%, 8.08 min → ~3.17 min)	1 notebook
Output cells cleared (stale results)	7 notebooks
No changes needed	7 notebooks

Verification

After fixes, the following checks all pass (zero matches):

- No `uniform_allocation` calls in any notebook
- No `P0_uniform` references in any notebook
- No `P0 (Uniform)` labels in any notebook
- No stale `8.08` min metrics in any output
- No stale `64.2%` / `0.644` coverage in any output
- All P0 references use “Spatial Baseline” or “Spatially-stratified” terminology
- All code uses `spatially_stratified_allocation` for P0 policy generation